

EXPERIMENT NO. 5

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AIM:

To create a Flask application that demonstrates template rendering by dynamically generating HTML content using the `render_template()` function.

PROBLEM STATEMENT:

Develop a Flask application that includes:

1. A homepage route (/) displaying a welcome message with links to additional pages.
2. A dynamic route (/user/<username>) that renders an HTML template with a personalized greeting.
3. Use Jinja2 templating features, such as variables and control structures, to enhance the templates.

Theory:-

♦ What Does the `render_template()` Function Do in a Flask Application?

The `render_template()` function in Flask is used to render HTML templates stored in the `templates` folder. Instead of returning plain text from a route, Flask can return a complete HTML page using this function.

Key Features of `render_template()`:

It loads an HTML file and returns it as the HTTP response.
It allows passing dynamic content (variables) to the template.
It supports template inheritance, which helps in maintaining reusable layouts.
It uses the Jinja2 templating engine to enable dynamic content rendering.

♦ What Is the Significance of the `templates` Folder in a Flask Project?

The `templates` folder is a default directory in Flask where all HTML files (templates) are stored. Flask automatically looks for templates in this folder when using `render_template()`.

Why Is the `templates` Folder Important?

It promotes separation of concerns by keeping HTML files separate from Python code, following the MVC (Model-View-Controller) architecture.
It provides an organized structure, helping to manage multiple pages and layouts efficiently.
Flask automatically searches for HTML templates inside this folder.
It supports template inheritance, which allows reusing layouts using a base template.

♦ What Is Jinja2, and How Does It Integrate with Flask?

Jinja2 is a powerful templating engine used in Flask to dynamically generate HTML content. It allows embedding Python-like expressions inside HTML templates.

♦ Key Features of Jinja2 in Flask

Jinja2 supports template variables using `{{ variable }}` syntax for dynamic content.
It allows control structures like loops (`{% for %}`) and conditionals (`{% if %}`).
It supports template inheritance for reusable layouts.
It provides filters and macros to modify output.

♦ How Jinja2 Integrates with Flask

Flask uses Jinja2 internally to process templates.
When calling `render_template()`, Flask renders the HTML template using Jinja2.
Developers can pass Python variables, lists, and objects to templates dynamically.

Github Link:- <https://github.com/MANO-RATH/WEBXEXP-.git>

Input:-

Style.css

```
/* Modern CSS Reset */
* {
  margin: 0;
  padding: 0;
  box-sizing: border-box;
}

/* Variables */
:root {
  --primary-color: #2563eb;
  --secondary-color: #1e40af;
  --background-color: #f8fafc;
  --text-color: #1e293b;
  --card-bg: #ffffff;
  --shadow: 0 4px 6px -1px rgb(0 0 0 / 0.1), 0 2px 4px -2px rgb(0 0 0 / 0.1);
}

/* Base Styles */
body {
  font-family: 'Inter', system-ui, -apple-system, sans-serif;
  background-color: var(--background-color);
  color: var(--text-color);
  line-height: 1.5;
  min-height: 100vh;
}

/* Container */
.container {
  max-width: 1200px;
  margin: 0 auto;
  padding: 2rem;
}

/* Navigation */
.nav {
  background-color: var(--card-bg);
  box-shadow: var(--shadow);
  padding: 1rem 2rem;
  margin-bottom: 2rem;
}

.nav-links {
```

```
    display: flex;
    gap: 1.5rem;
    list-style: none;
  }

/* Cards */
.card {
  background-color: var(--card-bg);
  border-radius: 0.5rem;
  padding: 2rem;
  box-shadow: var(--shadow);
  margin-bottom: 2rem;
}

/* Typography */
h1 {
  font-size: 2.5rem;
  font-weight: 700;
  margin-bottom: 1rem;
  color: var(--primary-color);
}

h2 {
  font-size: 1.875rem;
  font-weight: 600;
  margin-bottom: 1rem;
}

p {
  margin-bottom: 1rem;
  font-size: 1.125rem;
}

/* Buttons */
.button {
  display: inline-block;
  background-color: var(--primary-color);
  color: white;
  padding: 0.75rem 1.5rem;
  border-radius: 0.375rem;
  text-decoration: none;
  font-weight: 500;
  transition: all 0.2s;
}

.button:hover {
```

```
    background-color: var(--secondary-color);
    transform: translateY(-1px);
}

/* Lists */
ul {
    list-style-position: inside;
    margin-bottom: 1.5rem;
}

li {
    margin-bottom: 0.5rem;
}

/* Grid Layout */
.grid {
    display: grid;
    gap: 2rem;
    grid-template-columns: repeat(auto-fit, minmax(300px, 1fr));
}

/* Contact Info */
.contact-info {
    display: flex;
    flex-direction: column;
    gap: 1rem;
}

.contact-item {
    display: flex;
    align-items: center;
    gap: 0.5rem;
}

/* Responsive Design */
@media (max-width: 768px) {
    .container {
        padding: 1rem;
    }

    h1 {
        font-size: 2rem;
    }

    .nav-links {
        flex-direction: column;
    }
}
```

```

        gap: 1rem;
    }
}

```

About.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>{{ title }}</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
    <nav class="nav">
        <ul class="nav-links">
            <li><a href="{{ url_for('home') }}" class="button">Home</a></li>
            <li><a href="{{ url_for('user', username='Guest') }}" class="button">Profile</a></li>
            <li><a href="{{ url_for('contact') }}" class="button">Contact</a></li>
        </ul>
    </nav>

    <div class="container">
        <div class="card">
            <h1>{{ heading }}</h1>
            <p>{{ description }}</p>

            <div class="grid">
                <div class="card">
                    <h2>Our Mission</h2>
                    <p>We strive to create user-friendly web applications that enhance digital
experiences. Our goal is to simplify development with intuitive and scalable solutions.</p>
                </div>

                <div class="card">
                    <h2>Our Values</h2>
                    <ul>
                        {% for value in values %}
                        <li>{{ value }}</li>
                        {% endfor %}
                    </ul>
                </div>

                <div class="card">
                    <h2>Our Team</h2>
                    <p>Our team consists of passionate developers, designers, and strategists dedicated
to delivering high-quality solutions.</p>
                </div>
            </div>
        </div>
    </div>

```

```

        </div>
    </div>

    <a href="{{ url_for('home') }}" class="button">Back to Home</a>
</div>
</div>
</body>
</html>
Contact.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Get in Touch</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
    <nav class="nav">
        <ul class="nav-links">
            <li><a href="{{ url_for('home') }}" class="button">Home</a></li>
            <li><a href="{{ url_for('user', username='Guest') }}" class="button">Profile</a></li>
            <li><a href="{{ url_for('about') }}" class="button">About</a></li>
        </ul>
    </nav>

    <div class="container">
        <div class="card">
            <h1>Connect With Us</h1>
            <p>We're here to help! Reach out to us through any of the following channels.</p>

            <div class="card">
                <div class="contact-info">
                    <div class="contact-item">
                        <strong>Email:</strong>
                        <span><a href="mailto:{{ contact_info.email }}" class="button">{{
contact_info.email }}</a></span>
                    </div>
                    <div class="contact-item">
                        <strong>Phone:</strong>
                        <span>{{ contact_info.phone }}</span>
                    </div>
                    <div class="contact-item">
                        <strong>Location:</strong>
                        <span>{{ contact_info.address }}</span>
                    </div>
                </div>
            </div>
        </div>
    </div>

```

```

    </div>
</div>

<div class="card">
    <h2>Office Hours</h2>
    <p>We're available during the following times:</p>
    <ul>
        <li>Monday - Friday: 9:00 AM - 5:00 PM</li>
        <li>Saturday: 10:00 AM - 2:00 PM</li>
        <li>Sunday: Closed</li>
    </ul>
</div>

<div class="card">
    <h2>Quick Response</h2>
    <p>We typically respond to all inquiries within 24 hours during business days.</p>
</div>

    <a href="{{ url_for('home') }}" class="button">Return to Home</a>
</div>
</div>
</body>
</html>
Index.html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Modern Flask App</title>
    <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
    <nav class="nav">
        <ul class="nav-links">
            <li><a href="{{ url_for('home') }}" class="button">Home</a></li>
            <li><a href="{{ url_for('user', username='Guest') }}" class="button">Profile</a></li>
            <li><a href="{{ url_for('about') }}" class="button">About</a></li>
            <li><a href="{{ url_for('contact') }}" class="button">Contact</a></li>
        </ul>
    </nav>

    <div class="container">
        <div class="card">
            <h1>Welcome to Our Platform</h1>
            <p>Discover a modern web experience built with Flask and Jinja2 templating.</p>

```



```

<div class="grid">
  <div class="card">
    <h2>User Profiles</h2>
    <p>View personalized user profiles with dynamic content.</p>
    <a href="{{ url_for('user', username='Guest') }}" class="button">View Profile</a>
  </div>

  <div class="card">
    <h2>About Us</h2>
    <p>Learn more about our mission and values.</p>
    <a href="{{ url_for('about') }}" class="button">Learn More</a>
  </div>

  <div class="card">
    <h2>Contact</h2>
    <p>Get in touch with our team.</p>
    <a href="{{ url_for('contact') }}" class="button">Contact Us</a>
  </div>
</div>
</div>
</body>
</html>

```

User.html

```

<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>{{ username }}'s Profile</title>
  <link rel="stylesheet" href="{{ url_for('static', filename='style.css') }}">
</head>
<body>
  <nav class="nav">
    <ul class="nav-links">
      <li><a href="{{ url_for('home') }}" class="button">Home</a></li>
      <li><a href="{{ url_for('about') }}" class="button">About</a></li>
      <li><a href="{{ url_for('contact') }}" class="button">Contact</a></li>
    </ul>
  </nav>

  <div class="container">
    <div class="card">
      <h1>{{ username }}'s Profile</h1>

```

```

        {% if username == "Guest" %}
        <div class="card">
            <h2>Welcome Guest!</h2>
            <p>Consider creating an account to unlock personalized features and content.</p>
        </div>
        {% else %}
        <div class="card">
            <h2>Welcome Back!</h2>
            <p>We're glad to see you again, {{ username }}! Here's your personalized
dashboard.</p>
        </div>
        {% endif %}

        <div class="card">
            <h2>Your Interests</h2>
            {% if interests %}
            <ul>
                {% for interest in interests %}
                <li>{{ interest }}</li>
                {% endfor %}
            </ul>
            {% else %}
            <p>No interests found. Update your profile to add some!</p>
            {% endif %}
        </div>

        <a href="{{ url_for('home') }}" class="button">Back to Home</a>
    </div>
</div>
</body>
</html>

```

App.py

```
from flask import Flask, render_template
```

```
app = Flask(__name__)
```

```
@app.route('/')
def home():
```

```
    return render_template('index.html')
```

```
@app.route('/user/<username>')
```

```
def user(username):
```

```
    user_interests = {
```

```
        "Tejas": ["Coding", "Music", "Travel"],
```

```
        "Guest": [],
```

```
        "Alice": ["Photography", "Gaming", "Reading"]
```

```

    }
    interests = user_interests.get(username, ["No specific interests"])
    return render_template('user.html', username=username, interests=interests)

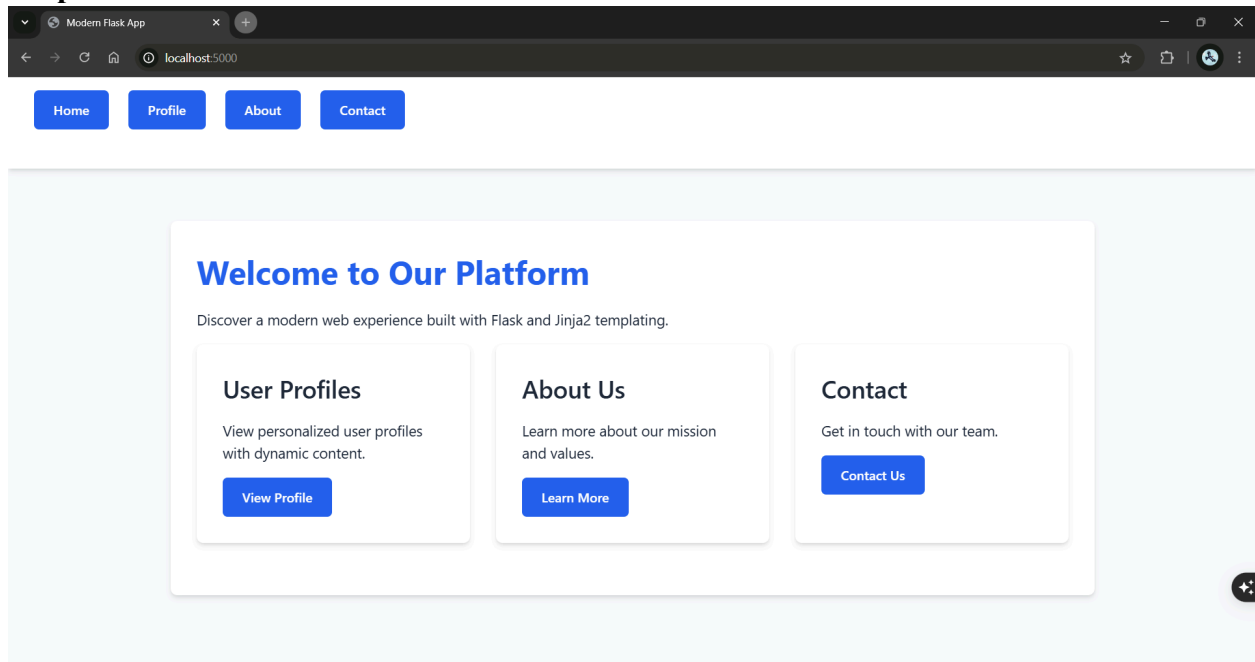
@app.route('/about')
def about():
    values = ["Innovation", "Integrity", "Collaboration", "Excellence", "Customer Focus"]
    return render_template('about.html',
        title="About Us",
        heading="About Us",
        description="This is a simple Flask application demonstrating template rendering.",
        values=values)

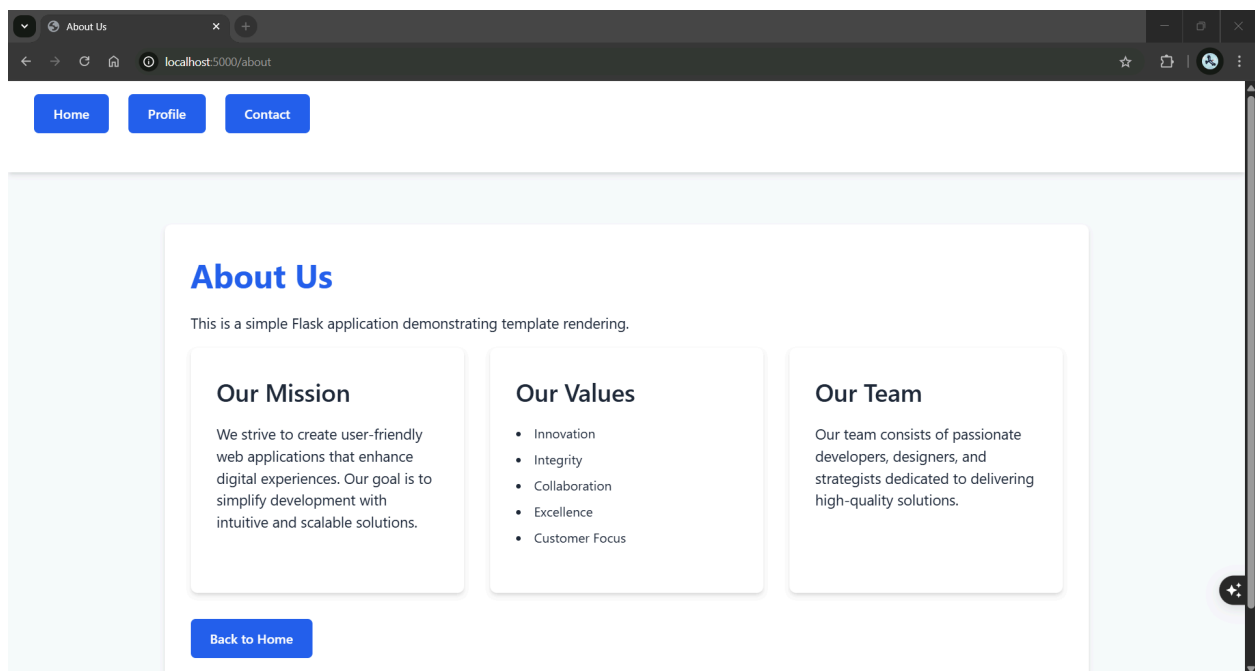
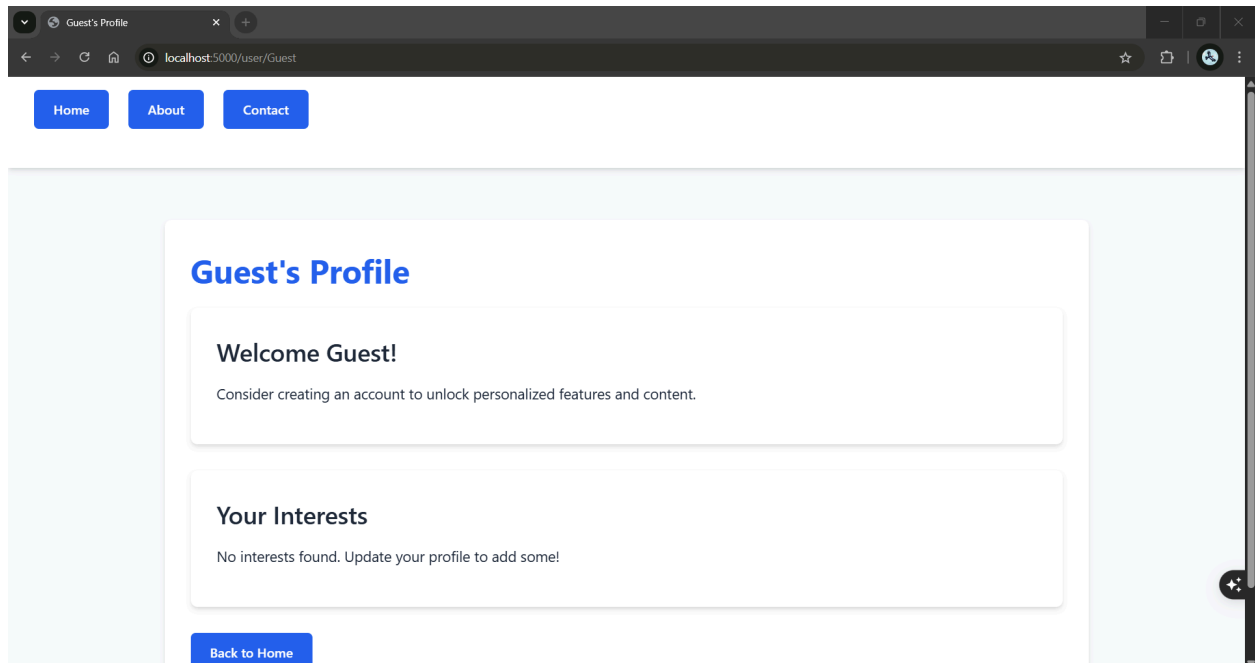
@app.route('/contact')
def contact():
    contact_info = {
        "email": "manorathital27@gmail.com",
        "phone": "+123 456 7890",
        "address": "Kalyan, Maharashtra"
    }
    return render_template('contact.html', contact_info=contact_info)

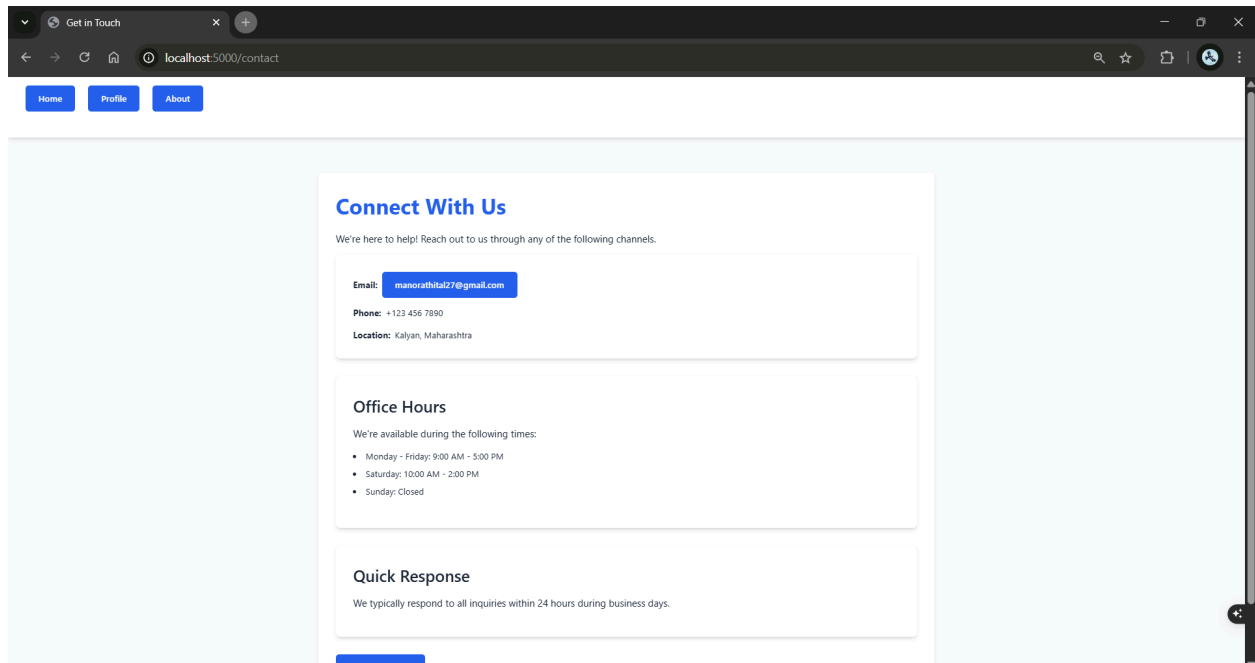
if __name__ == '__main__':
    app.run(debug=True)

```

Output:-







◆ Conclusion:

The Flask application successfully demonstrates template rendering using the `render_template()` function and Jinja2 templating features. The homepage provides navigation links, while the dynamic user route personalizes content by passing data to an HTML template. By utilizing variables and control structures, the application efficiently generates dynamic web pages, showcasing Flask's ability to create flexible and interactive web applications.